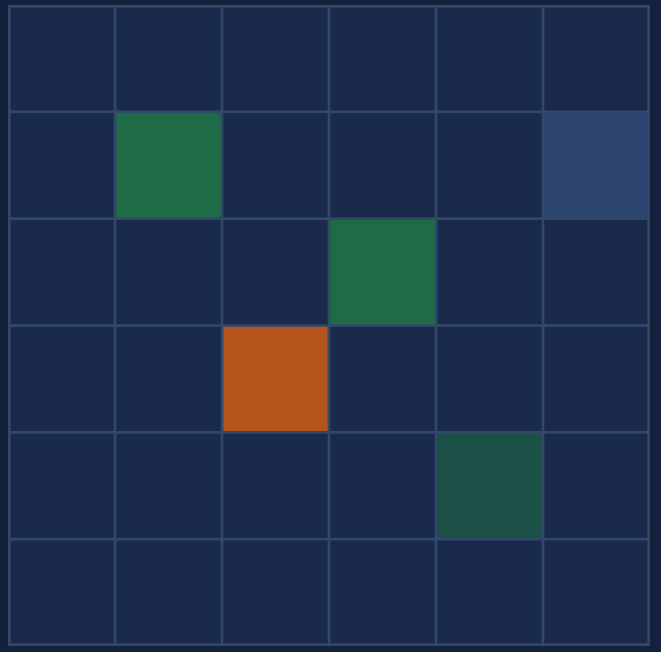
INF131 · TECHNOLOGY AND AI · WEEK 4

Data quality and cleaning

Rows, columns, and how to turn messy data into evidence you can trust.



WHERE WE ARE

From thinking in steps to clean data

WEEK 3

Computational thinking

Break a problem into steps, conditions and repetition.

TODAY

Data quality

Cleaning is the same idea: a fixed sequence of steps applied to every row.

LAB

Power Query

Records each step, so the whole recipe can run again on new data.

A clean table is the input to everything that follows: pivot tables, SQL, Power BI and AI.

TODAY

Three parts, two short breaks

Time	Part	What we cover
0:00 – 0:50	1 · Data and tables	Rows, columns, tidy data, data types
0:50 – 1:00	Break	
1:00 – 1:50	2 · Dirty data	Missing values, duplicates, spelling, formats, outliers
1:50 – 2:00	Break	
2:00 – 2:50	3 · Cleaning with a method	A workflow, Excel fixes, Power Query and the lab

Bring a question from Part 2 into the lab: every problem we name today, you will fix by hand in Power Query.

What we mean by data

Data is a record of something that happened, written down so a computer can count it.

Each event below becomes **one row** in a table. Each detail becomes **one column**.

A sale in a shop

date · branch · product · units · price

A country in a year

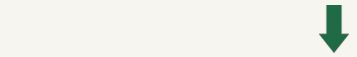
country · year · GDP per capita · inflation

A bank loan

customer ID · age · income · amount · term

Anatomy of an Excel sheet

Column C



	A	B	C	D	E
1	Date	Branch	Product	Units	Price (ALL)
2	03/10/2026	Tirana	Coffee beans	12	1,200
3	03/10/2026	Durrës	Green tea	5	650
4	04/10/2026	Shkodër	Coffee beans	8	1,200
5	04/10/2026	Tirana	Olive oil	3	1,500
6	05/10/2026	Vlorë	Green tea	10	650
7	05/10/2026	Durrës	Olive oil	6	1,500

Row 4 = one record

One sale: 8 bags of coffee beans in Shkodër on 4 October.

Column C = one variable

The same kind of fact for every row: which product was sold.

Cell C4 = one value

Address = column letter + row number. Row 1 holds names, not data.

One idea, three vocabularies

Excel	Databases	Statistics and economics
Row	Record	Observation
Column	Field	Variable
Cell	Value	Data point
Sheet or table	Table	Dataset

Rule of thumb: every row in a table describes the same kind of thing — one sale, one country-year, one customer.

PART 1 · DATA AND TABLES

Three rules of tidy data

1

Each variable is a column

2

Each observation is a row

3

Each value is a cell

WHAT BREAKS THE RULES

Merged cells · subtotal rows in the middle of the data · two facts in one cell (“Tirana – 12 units”) · notes typed into data cells · colour used as information

Same data, two shapes: wide and long

WIDE · one column per month

Branch	Oct	Nov	Dec
Tirana	120	135	160
Durrës	80	90	110



Unpivot

LONG · one row per branch-month

Branch	Month	Units
Tirana	Oct	120
Tirana	Nov	135
Tirana	Dec	160
Durrës	Oct	80
Durrës	Nov	90
Durrës	Dec	110

World Bank downloads arrive wide, with one column per year. Pivot tables and Power BI prefer long. Power Query's **Unpivot** converts one into the other in a single step.

One column, one data type

TEXT
Tirana
Branch, Product

NUMBER
12 · 1200
Units, Price

DATE
03/10/2026
Date of sale

TRUE / FALSE
TRUE
Paid?

Excel decides cell by cell

One column can hide text among numbers. Clue: a number sitting on the left of its cell is stored as text.

Power Query decides per column

It sets one type for the whole column and flags every value that doesn’t fit as an error.

02

What makes data dirty

Missing values · duplicates · inconsistent categories · wrong types and formats · impossible values

PART 2 · DIRTY DATA

When dirty data gets expensive

2013 · ECONOMICS

A debt study’s missing countries

A famous paper linked high public debt to slow growth. A student replicating it found an average that skipped several countries. The headline result weakened.

2020 · PUBLIC HEALTH, UK

Lost test results

Almost 16,000 positive COVID-19 tests were left out of England’s daily figures. The data passed through the old .xls format, which stops at 65,536 rows.

2012 · BANKING

The “London Whale”

A JPMorgan risk model ran on spreadsheets fed by copy-paste, and a formula error understated the risk. Trading losses reached about $6 billion.

None of these errors was clever: a cell range, a file format, a copy-paste. Ordinary checks would have caught them.

Six questions to ask of any dataset

Accuracy

Are the values correct?

Completeness

Is anything missing that should be there?

Consistency

Is the same thing written the same way?

Validity

Do values follow the rules? Units ≥ 0, dates are real dates.

Uniqueness

Is each record there only once?

Timeliness

Is it recent enough for the question?

PROBLEM 1 · COMPLETENESS

Missing values

Missing data hides in many forms:

blank .. N/A - 0 9999

World Bank downloads mark missing values with “..”

Leave it empty and report how many are missing.

Remove the row if only a few rows are missing, at random.

Fill it only with a rule you can state and defend.

Units: 10, 20, *blank*

AVERAGE = 15

Units: 10, 20, 0

AVERAGE = 10

Zero is a value. Blank means “we don’t know”.

PROBLEM 2 · UNIQUENESS

Duplicates

Branch	Product	Units	Price	Revenue
Tirana	Coffee beans	12	1,200	14,400
Durrës	Green tea	5	650	3,250
Tirana	Coffee beans	12	1,200	14,400
Vlorë	Green tea	10	650	6,500

Total with the repeat

38,550 ALL

True total

24,150 ALL

Exact duplicates

Every cell identical. Remove Duplicates handles them.

Near duplicates

“tirana” vs “Tirana”, two date formats. Clean the text first, then remove duplicates.

Not every repeat is an error

Two customers can buy the same thing on the same day. A transaction ID tells them apart.

PROBLEM 3 · CONSISTENCY

Inconsistent spelling and categories

PIVOT · UNITS BY BRANCH · BEFORE

Branch as typed	Units
Tirana	12
tirana	5
TIRANA	3
Tiranë	7
“Tirana ” (space at end)	4



AFTER CLEANING

Branch	Units
Tirana	31

Five rows for one city. The real total, 31, appears nowhere in the “before” table.

TRIM removes extra spaces

PROPER fixes capitals

Replace or a lookup list: Tiranë → Tirana, ALB → Albania

PROBLEM 4 · VALIDITY

Wrong types and formats

“1,200”

Numbers stored as text

SUM skips them without warning, so totals come out too low.

1.234,5 · 1,234.5

Decimal comma or point

Albanian settings use a comma for decimals, English files a point. The wrong setting changes the size of numbers.

03/10/2026

Day or month first?

3 October or 10 March? Store real dates; how they look is a separate choice.

Also check **units**: thousands or millions? lek or euro? Put the unit in the column name, for example Price (ALL) or GDP (current US$).

PROBLEM 5 · ACCURACY

Impossible and suspicious values

Impossible

Units = −3

Date = 31/02/2026

Age = 250

Break a rule, so they must be wrong. Fix or remove them, and write down what you did.

Suspicious

Green tea price = 65,000

(usually 650)

A typo, a unit mix-up, or a real event? Check the source before you change anything.

How to find them: Sort | MIN and MAX | Filter | Conditional formatting | Column profile in Power Query

Investigate before you delete. An outlier can be an error — or the most important row in the table.

EXERCISE

Find the problems

Find at least six. For each one: what is wrong, and how would you fix it?

	A · Date	B · Branch	C · Product	D · Units	E · Price (ALL)
2	03/10/2026	Tirana	Coffee beans	12	1,200
3	2026-10-03	tirana	coffee beans	12	1200
4	04/10/2026	Shkoder	Coffee Beans		1,200
5	04/10/2026	Tiranë	Olive oil	-3	1,500
6	05/10/2026	Vlorë	Green tea	10	65000
7	05/10/2026	Durres	Olive oil	six	1,500
8	31/02/2026	Vlorë	Green Tea	4	650

03

Cleaning with a method

A workflow you can repeat · quick fixes in Excel · Power Query · today’s lab

A cleaning workflow in four steps

1

Keep the raw file

Save the original untouched. Work on a copy.

2

Inspect

Sort, filter, count blanks, check MIN and MAX. List the problems.

3

Clean

Fix one problem at a time: text first, then duplicates, then types.

4

Check and document

Compare row counts and totals before and after. Write down every step.

Same raw data + same steps = same clean table. That is what makes your work reproducible, and checkable by a lecturer, a manager or an auditor.

PART 3 · CLEANING WITH A METHOD

Each problem has a tool

Problem	Quick fix in Excel	Step in Power Query
Extra spaces	TRIM( )	Transform › Format › Trim
Mixed capitals	PROPER( ), UPPER( )	Transform › Format › Capitalize Each Word
Missing as “..”	Find and Replace	Replace Values: “..” → null
Blank rows	Filter, COUNTBLANK( )	Home › Remove Rows › Remove Blank Rows
Duplicates	Data › Remove Duplicates	Home › Remove Rows › Remove Duplicates
Numbers as text	VALUE( ), Text to Columns	Change the column type to Decimal Number

An Excel fix changes the cells once. A Power Query step is saved and runs again every time.

PART 3 · CLEANING WITH A METHOD

Power Query: cleaning as a recipe



New data next month? Click **Refresh** and every step runs again: week 3's algorithm, saved.

LAB · POWER QUERY WITH WORLD BANK DATA

Today’s lab, step by step

STEP 1

Import

Data › Get Data › From File. Open the World Bank file and choose Transform Data.

STEP 2

Headers

Use First Row as Headers, if Power Query hasn’t done it already.

STEP 3

Missing values

Replace “..” with null, so Excel treats it as empty.

STEP 4

Blanks and repeats

Remove Blank Rows, then Remove Duplicates.

STEP 5

Data types

Text for country, whole number for year, decimal number for values.

STEP 6

Load and check

Close and Load to a new sheet. Compare the row count with the raw file.

☆ **Extension:** Unpivot the year columns so each row is one country in one year.

AI can help you clean. You still check.

Ask AI to…

- Write a formula: “Excel formula to trim spaces and capitalise the names in column B”
- Explain a Power Query step you don’t understand
- Suggest which problems to look for in a column

Then you…

- Compare row counts and totals before and after
- Never paste personal or confidential data into an AI tool
- Record the tool and what it did in your AI-use statement

TO REMEMBER

Four things to take into the lab

1. One row = one observation, one column = one variable, one cell = one value.
2. Blank is not zero. Decide how to treat missing data, and say so.
3. Clean text before removing duplicates. Check totals after.
4. Never edit the raw file. Record your steps so they can run again.

PROJECT

Cleaned dataset with documented Power Query steps, due week 7

READING

Microsoft Learn: Power Query, intro module

NEXT WEEK

Analysis and visualisation: pivot tables and charts